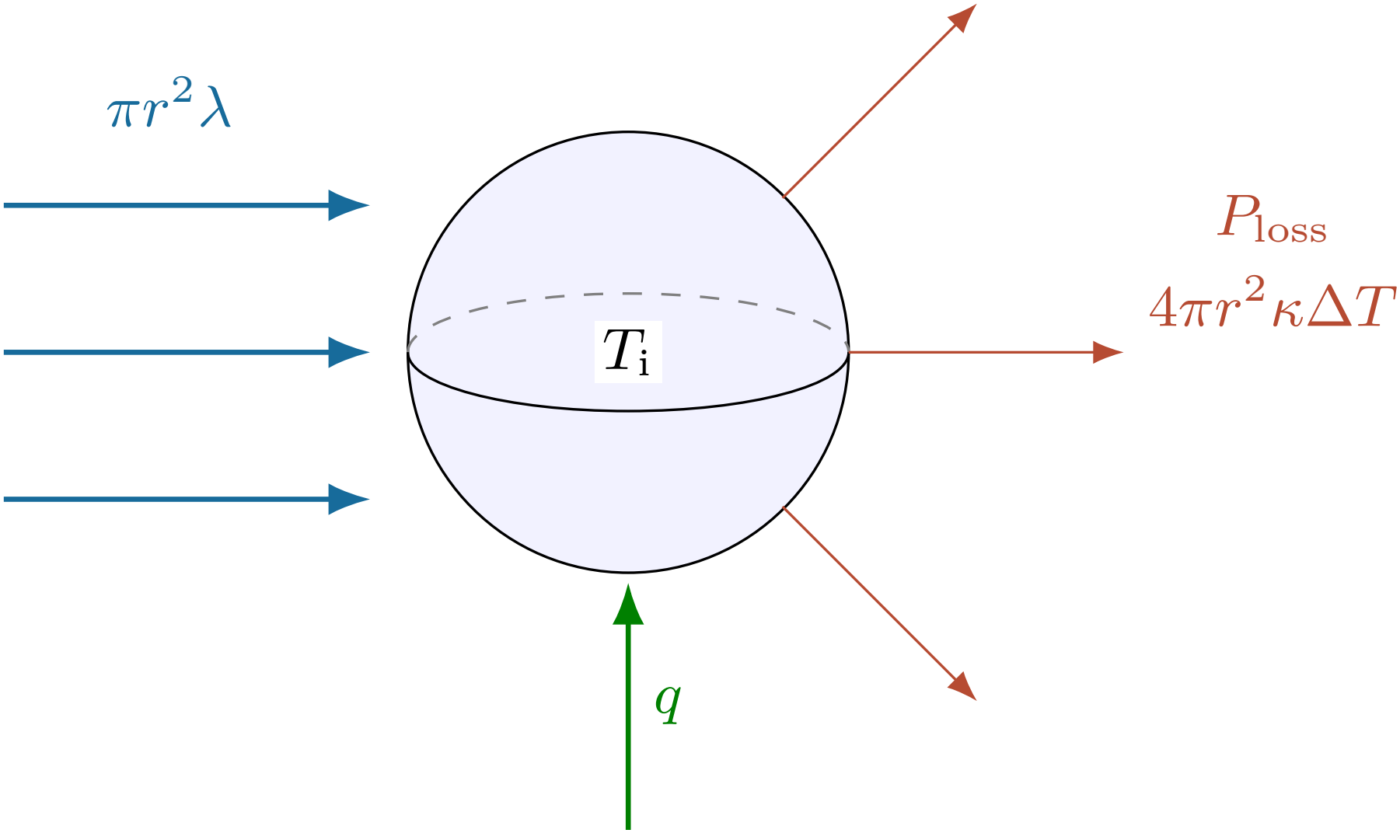

$$q + P_\mathrm{sun} = P_\mathrm{loss}$$